WORKED SOLUTION

STEP II 1998, Question 11

Mechanics • Projectile Motion

STEP II 1998, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/98-s2-q11

This question has a full guided treatment in *Beyond STEP*, Section 11.11, *How Fielders Catch*: https://beyondstep.keithmcnulty.org/ch11/fielders.html

The key idea

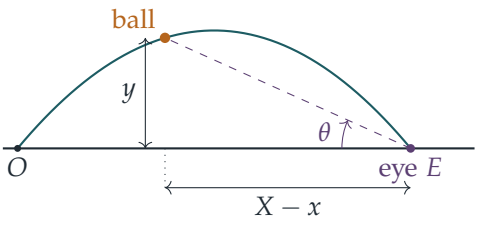
Put the origin at the point where the ball is struck, at eye level, and write $\tan\theta$ as "height of the ball over horizontal distance from the fielder's eye". The height of a projectile that starts and finishes at eye level factorises as $\frac{1}{2}g\,t(T-t)$, where T is the time of flight, and if the fielder ends up in the right place the horizontal gap is also a multiple of $T-t$. The factor $T-t$ cancels, leaving $\tan\theta$ proportional to t.

Solution

We model the ball as a particle moving under gravity alone (no air resistance), and the fielder's eye as a point at the same height as the point where the ball is struck. Take x horizontally from the striking point O towards the fielder and y vertically upwards. Let the ball's initial velocity have horizontal component $U>0$ and vertical component $W>0$. Then at time t

$$x = Ut, \qquad y = Wt - \tfrac{1}{2}gt^2 = \tfrac{1}{2}g\,t\,(T-t), \qquad \text{where } T = \frac{2W}{g}$$

is the time at which the ball returns to eye level.



If the fielder's eye is at horizontal position X (with $X > x$ while the ball is in flight), the line of sight rises at angle θ above the horizontal, where

$$\tan\theta = \frac{y}{X - x}.$$

The stationary fielder

"Perfectly placed" means the ball arrives at her eye: her eye is at the point where the ball returns to eye level, $X = UT$. Then for $0 < t < T$

$$\tan\theta = \frac{\frac{1}{2}gt(T-t)}{UT - Ut} = \frac{\frac{1}{2}gt(T-t)}{U(T-t)} = \frac{g}{2U}t,$$

cancelling $T - t$, which is non-zero while the ball is in flight. Hence

$$\frac{\mathrm{d}}{\mathrm{d}t}(\tan\theta) = \frac{g}{2U},$$

which is constant, as required. (Just before the catch, $\tan\theta \to gT/(2U) = W/U$: she sees the ball coming in at exactly the angle at which it was struck, as symmetry of the parabola suggests.)

The running fielder

Now let the next ball have velocity components U, W (so again $y = \frac{1}{2}gt(T-t)$ with $T = 2W/g$), and let the fielder start at distance D from O at the moment the ball is struck, running towards the batsman at constant speed v. Her eye is at $X = D - vt$, so

$$\tan\theta = \frac{\frac{1}{2}gt(T-t)}{D - (U+v)t}. \qquad (*)$$

She catches the ball exactly when her eye is where the ball returns to eye level at time T, that is, when

$$D - vT = UT \quad \Longleftrightarrow \quad D = (U+v)T.$$

We show that this happens if and only if $\frac{\mathrm{d}}{\mathrm{d}t}(\tan\theta)$ is constant, so "choose v to keep $\frac{\mathrm{d}}{\mathrm{d}t}(\tan\theta)$ constant" is the right rule.

If she has chosen the right speed, $D = (U+v)T$, and $(*)$ becomes

$$\tan\theta = \frac{\frac{1}{2}gt(T-t)}{(U+v)(T-t)} = \frac{g}{2(U+v)}t, \qquad \text{so} \qquad \frac{\mathrm{d}}{\mathrm{d}t}(\tan\theta) = \frac{g}{2(U+v)},$$

a constant, exactly as in the first part with U replaced by the ball's speed $U + v$ relative to her.

Ask yourself Isn't that only half of what's needed? Could some other v also give a constant rate?

No. Suppose $\frac{\mathrm{d}}{\mathrm{d}t}(\tan\theta) = k$, a constant, throughout the flight. At $t = 0$ the ball is at eye level, so $\tan\theta = 0$, and hence $\tan\theta = kt$. Putting this into $(*)$ and dividing by t (for $0 < t < T$),

$$\tfrac{1}{2}g\,(T - t) = k\big(D - (U + v)t\big) \quad \text{for all } t \in (0, T).$$

Two linear functions of t that agree on an interval have equal coefficients, so

$$\tfrac{1}{2}g = k(U + v) \qquad \text{and} \qquad \tfrac{1}{2}gT = kD.$$

The first gives $k \neq 0$; dividing the second by the first gives $D = (U + v)T$, which is precisely the catching condition.

So the speed $v = D/T - U$ that brings her to the ball is exactly the one for which $\frac{\mathrm{d}}{\mathrm{d}t}(\tan\theta)$ stays constant. In practice she cannot compute $D/T - U$ while the ball is in the air, but she can watch: if the ball seems to rise ever faster in her view she is too far back and should run faster, and if it seems to rise ever slower she should ease off. (Since she runs *towards* the batsman, the ball falls short of her starting point, $D > UT$, so $v > 0$.)

Checks

- Setting $v = 0$ in the second part recovers the first part: $D = UT$ and rate $g/(2U)$.
- Units: g/U has units $\mathrm{m\,s^{-2}}/\mathrm{m\,s^{-1}} = \mathrm{s^{-1}}$, right for the rate of change of a pure number.
- A numerical example: struck at 20 $\mathrm{m\,s^{-1}}$ at 45° with $g = 9.8\ \mathrm{m\,s^{-2}}$, $T \approx 2.886$ s and the range is 40.82 m; a fielder standing there sees $\tan\theta$ grow at $g/(2U) \approx 0.346\ \mathrm{s^{-1}}$. If she instead starts 49.47 m away and runs at 3 $\mathrm{m\,s^{-1}}$, the rate is $0.286\ \mathrm{s^{-1}}$. Differentiating $(*)$ numerically at many times confirms both rates are constant, and that a wrong v gives a rate that changes.

Common slips

- Measuring θ from the batsman rather than from the fielder's eye: the horizontal distance in $\tan\theta$ is $X - x$, not x.
- Proving only that the right v gives a constant rate, without showing that a constant rate forces the right v; it is the converse that makes this a usable rule.

Beyond STEP

This is the "optical acceleration cancellation" idea from the psychology of catching: fielders seem to run so that the ball's image rises at a steady rate, without ever predicting where it will land. It is an example of a feedback control law, where an observed quantity is held fixed rather than a trajectory being computed. For projectile motion, relative velocity and the beginnings of this way of thinking, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

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